



Gnoming in Space

Gabe Newell is launching a garden gnome into space, as thanks for New Zealand's hospitality. <https://digit.in/dec20-29>



Chang'e-5 Mission

China's ambitious mission could bring back some lunar fresh samples, after a 44-year hiatus. <https://digit.in/dec20-30>

sample return missions from bodies thought to have no life such as Venus, an assumption that may be wrong. The restricted category requires the most stringent handling of samples, from say potential missions to Mars.

The space agencies and private spaceflight companies in the spacefaring nations have to meet the requirements of the treaty on their own, and thus have to develop regulations and guidelines to meet the requirements of the treaty. NASA for example has its own Planetary Protection Policies which have detailed outlines for the decontamination of spacecraft. The objective of these policies is to ensure that any spacecraft going on interplanetary visits carries an appropriate bioburden to prevent contamination. The bioburden here refers to the amount of viable organisms piggybacking on the spacecraft, measured in units of surface area or volume. It is similar to the approaches used to measure how clean medical equipment is. The overall goal of the decontamination efforts of space agencies is to prevent organisms from Earth contaminating worlds such as Europa, Mars and the Moon. At the same time, it also means protecting the Earth from any microorganisms that can hitch a ride to the planet, on any future sample return missions.

POST-DETECTION POLICY

What should we do if ET calls us? That is, if we get a directed message towards the Earth from an advanced civilization, how are we supposed to respond to that? IAA-SETI, the group of scientists who transitioned from public funding to private funding to scan the skies for intelligent signals, has a set of protocols that have no regulatory power, and cannot be enforced as a part of any international law or multi nation treaty. However, these protocols have been agreed to by all

the organisations that are searching for extraterrestrial messages.

There are two parts of the protocols, one dealing with what should be done with the message when it is received, and the other dealing contains the recommendations for drafting and dispatching a suitable response. When the signal is detected, the entity that discovers the signal should first verify its authenticity and then inform other bodies conducting SETI so that it can be verified. Then, they should disclose the contents of the signal, making sure to point out that it is an unexplained phenomenon if they cannot verify that it has been sent by an advanced intelligence. If such a signal is detected, the protocol is



Image taken from the Viking II lander, one of the earliest interplanetary spacecraft to be decontaminated according to COSPAR guidelines.

that it should be widely informed to the general public and the scientific community, which means that no body participating in SETI can keep such a signal secret.

According to the protocols, after a message has been received, the United Nations General Assembly should determine if there should be a response from humans. An appropriate international body should be set up to figure out what the contents of the response should be. The message should represent all of humanity, and not just the interest of any single nation. As such, the response should represent a broad consensus between all the countries. Additionally, the

message should be made available to the general public, in advance of the transmission. Individual states are not to make contact directly, and should not support any efforts to make contact or transmit messages that do not follow the principles of the post-detection policy.

THE MILITARY APPROACH

There are no official guidelines for a military response, at least none that are public. UFOlogists, conspiracy theorists, and shady YouTube channels refer to a joint Navy-Army briefing in 1950, where how contact can be initiated by humans and how contact is likely to be initiated by aliens was allegedly outlined. There is no shred of evidence that this was ever an official military briefing, but the seven phases of contact is interesting nonetheless. In Phase I, the planet would be detected remotely using astronomical instruments. In Phase II, probes would be sent to survey the planet from orbit. In Phase III, manned vehicles would scout and finger their alien counterparts, to figure out the capabilities of the alien vehicles, gear and technologies. Then, only in Phase IV, will the humans set out to determine if the aliens are in fact hostile, and if further engagement is beneficial to us. Phase V is when scientific expeditions would land in remote areas to collect samples for study. The next step is Phase VI, where we reveal that we exist to the aliens, which can be achieved in various ways such as a signal or allowing detection of a spacecraft on their technologies. Finally in Phase VII, humans would initiate direct face to face contact with the extraterrestrials. According to UFOlogists, the alleged alien abductions that have occurred on Earth are because extraterrestrials are in Phase V of their contact with humans. Make no mistake, the scientific community considers the seven phases of contact to be complete hokum. 